



Performance and Reproducibility of Large Language Models in Named Entity Recognition: Considerations for the Use in Controlled Environments

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Abstract

Introduction Recent artificial intelligence (AI) advances can generate human-like responses to a wide range of queries, making them a useful tool for healthcare applications. Therefore, the potential use of large language models (LLMs) in controlled environments regarding efficacy, reproducibility, and operability will be of paramount interest.

Objective We investigated if and how GPT 3.5 and GPT 4 models can be directly used as a part of a GxP validated system and compared the performance of externally hosted GPT 3.5 and GPT 4 against LLMs, which can be hosted internally. We explored zero-shot LLM performance for named entity recognition (NER) and relation extraction tasks, investigated which LLM has the best zero-shot performance to be used potentially for generating training data proposals, evaluated the LLM performance of seven entities for medical NER in zero-shot experiments, selected one model for further performance improvement (few-shot and fine-tuning: Zephyr-7b-beta), and investigated how smaller open-source LLMs perform in contrast to GPT models and to a small fine-tuned T5 Base.

Methods We performed reproducibility experiments to evaluate if LLMs can be used in controlled environments and utilized guided generation to use the same prompt across multiple models. Few-shot learning and quantized low rank adapter (QLoRA) fine-tuning were applied to further improve LLM performance.

Results and Conclusion We demonstrated that zero-shot GPT 4 performance is comparable with a fine-tuned T5, and Zephyr performed better than zero-shot GPT 3.5, but the recognition of product combinations such as product event combination was significantly better by using a fine-tuned T5. Although Open AI launched recently GPT versions to improve the generation of consistent output, both GPT variants failed to demonstrate reproducible results. The lack of reproducibility together with limitations of external hosted systems to keep validated systems in a state of control may affect the use of closed and proprietary models in regulated environments. However, due to the good NER performance, we recommend using GPT for creating annotation proposals for training data as a basis for fine-tuning.

1 Introduction and Objective

Over the last few years artificial intelligence/machine learning (AI/ML) has been increasingly used throughout drug development and across therapeutic areas from drug discovery, non-clinical and clinical research, clinical trial management, and drug manufacturing to post marketing safety surveillance [1–8]. With the increased use of AI in

healthcare, ethical challenges become more apparent. In 2021 the WHO issued a report on AI in health with six guiding principles for its design and use to address ethical considerations [9]. Managing the permanently increasing volume, variety, and velocity of pharmacovigilance (PV) data in the framework of regulatory requirements presents a remarkable challenge. One important AI/ML benefit resides in its ability to learn from real-world use and experience, and the AI/ML capability for performance improvement. AI-based automation in PV and especially in individual case safety report (ICSR) processing represents a significant opportunity for efficiency improvement and cost reduction [10–12].

Recent publications are addressing quality aspects of AI/ML use as software as a medical device (SaMD) in

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Key Points

Reproducibility of generated text is a prerequisite when using LLMs in regulated environments.

Open-source LLMs ensure reproducibility and provide good results by fine-tuning, few-shot learning, and guided generation.

LLM zero-shot performance is a good starting point to support training-data generation.

health care focused on risk considerations [13–15] and the use of AI in the medicinal product life cycle [16]. Although requirements for validation largely remain the same, additional activities tailored to AI/ML systems are required to document evidence that those systems are fit for purpose [17]. US Food and Drug Administration (FDA) proposes to place AI/ML-based systems into one of four categories to reflect the risk associated with the clinical situation and the significance of information provided [13], and introduced a total product lifecycle (TPLC) approach for the regulation of AI/ML-based software products [18]. Beside common computer validation aspects, Kompa et al. described in a systematic review that a lot of AI/ML implementations do not reflect current characteristics of good machine learning practices and trends in machine learning [19]. Intensifying efforts to identify potential adverse drug reactions (ADRs), and subsequently create, store, analyze, and communicate results in individual case safety reports (ICSRs), could potentially produce more reports, but would not necessarily result in more rapid or accurate identification or understanding of true safety issues [20].

Besides classical computer validation aspects like result reproducibility, system performance, controlled state persistence, the investigation of which system, methodology, and appropriate training data are best for the intended AI/ML system use is of paramount importance.

In machine learning, the system evaluation analyzes how well a model performs its intended task. Accuracy may be insufficient in situations with imbalanced classes, which is mainly given for instance in adverse event (AE) detection in real-world datasets. In pharmacovigilance, model evaluation is therefore focusing on F1 score as the harmonic mean of precision and recall (i.e., with increased precision and maintained F1 score recall is decreased) and preferably on recall, since recall measures indicate how well the model finds all positive occurrences in a dataset.

A GxP validated system must be kept in a state of control. Therefore, it is important to establish appropriate measures for process and system maintenance to keep the validated state during routine operation. This ongoing process and system verification provides documented evidence that the system remains in a state of control during production use [21].

The system remains in a GxP validated state as long as conditions and control parameters remain unchanged. By using the current externally hosted OpenAI model family, this status can hardly be preserved due to the lack of direct control over system infrastructure, which may potentially affect system quality by provider's system or infrastructure changes.

Reproducibility is an essential ingredient of computer system validation, meant to verify the documented results and claims and to enable a continuous life cycle management process. AI methods and algorithms are increasingly used in critical systems or applications, where their decisions can have an impact on people and society, and their improper operation may cause harm. Therefore, ensuring reproducibility of algorithm results is a method to test their quality and a signal of their credibility [22].

Measuring the closeness of results under the same repeatability conditions (e.g., same sample, measurement procedure, operating conditions) establishes scientific evidence during validation that the process is reproducible and will consistently deliver the requested quality [23].

The validation protocol should specify a sufficient number of replicate process runs to demonstrate reproducibility and provide an accurate measure of variability among successive runs [24].

In our study we investigated the performance (partial F1 score and partial recall) of several externally hosted OpenAI models versus internally hosted large language models (LLMs) to explore, by example of AE detection tasks in pharmacovigilance, if pre-trained or fine-tuned smaller LLMs would reach the efficacy of OpenAI models like GPT 4. We considered NLP tasks that go beyond named-entity recognition (NER) (e.g., detection of a drug or dose) such as linking two entities with each other (e.g., extracting product dose combinations). Recent advances in text-generating models make it possible to cast many tasks relevant for the medical domain as text-generation tasks to solve multiple use-cases with a unified approach. We emphasize here named-entity recognition and linking, as they are important applications with many downstream use-cases such as detection of adverse events or entity-based search.

As a baseline model, we used in our experiments T5-Base since it was shown that T5 outperformed other AE detection models based on bidirectional encoder representations from transformers (BERT) variants and can be used to apply the same model for multiple and diverse datasets [25, 26]. In addition, we investigated the reproducibility of OpenAI models as since November 2023 OpenAI introduced new methods to ensure better reproducibility of their models.

A recent publication by Wadhwa et al. [27] demonstrated that relation extraction on the adverse drug effect (ADE) corpus [28] and non-medical datasets can be achieved using GPT-3 and few-shot learning, Hu et al. [29] showed good results for named entity recognition on data from vaccine adverse event reporting system (VAERS) using GPT 3.5 and 4 with prompt engineering and few-shot learning. Li et al. [30] focused on LLMs to extract adverse events from surveillance reports on influenza vaccine adverse events from VAERS.

Since those publications have already demonstrated that current LLMs can deliver results that match with or improve previous methods, we do not focus in our paper on further improving model performance, but

- to investigate if and how GPT 3.5¹ and GPT 4 models can be directly used as a part of a GxP-validated system,
- to investigate if the expected better performance of GPT 4 versus GPT 3.5 justifies the higher costs and longer execution time per request,
- to investigate the LLM zero-shot NER performance to find the best open-source model for further performance improvement steps (i.e., few-shot learning, fine-tuning),
- to investigate which LLM has the best zero-shot performance to be used potentially for generating training data proposals,
- to investigate how smaller open-source LLMs perform in contrast to GPT models and to a small fine-tuned T5 Base, since smaller open-source LLMs are easier to implement in local infrastructures, easier to maintain, and faster to train with less resources and costs,
- and to recommend a solution for LLMs for the use in a controlled environment.

¹ Please note that ‘T5’, ‘GPT 3.5’, and ‘GPT 4’ refer in our experiments to the models ‘T5-Base’, ‘GPT 3.5 Turbo’, and ‘GPT 4 Turbo’, respectively.

2 Methods

2.1 Overview

In our study we were using a fine-tuned T5 Base as a reference model which shows good performance in PV-related NER tasks [31, 32]. Our assumption was that all larger model sizes compared to T5 Base would typically result in better performance, but it would also require more computing resources to train and run. In our experiments we strove to find the best balance between efforts and performance depending on the results of our experiments.

LLMs are trained with large amounts of text data to determine the most likely continuation of an input text. The model proposes a list of the most probable subsequent words, and a text generation algorithm is employed to select them. Algorithms that do not consistently select the most likely word can enhance the overall probability of the complete output and generate more variable and natural text. Temperature is a common parameter, which regulates the probability of selecting the most likely next word. Its value typically starts at 0 and progress to higher values, which introduce greater randomness or creativity. In our study, the temperature parameter for all models was set to 0 to reduce the degree of randomness in response generation.

The term zero-shot refers to a prompting technique wherein the model is presented with a task without any additional examples, whereas in the case of a few-shot prompt, the prompt comprises multiple examples of the task and the anticipated outcome [33–35].

Experiments were executed between October 2023 and February 2024. We started with prompt engineering² of GPT 3.5 Turbo [33] (gpt-3.5-turbo-1106) and GPT 4 Turbo [36] (gpt-4-1106-preview) models [37]. Although GPT 3.5 and GPT 4 experiments were originally executed using Microsoft Azure, we repeated our experiments with Open AI's public API (version 1.6.1) to allow reproducibility for the research community. Results obtained by the OpenAI API Python SDK for text generation did not principally deviate from those executed in Azure.

Both GPT models return a maximum of 4096 output tokens³, while the context window of GPT 3.5 Turbo with 16,385 tokens is lower when compared with GPT 4 Turbo context window of 128,000 tokens. We performed qualitative prompt experiments independent of entity performance. We focused on modifying the system and user prompt to

² Prompt engineering is the process of writing, refining, and optimizing inputs to allow AI models to create specific, high-quality outputs.

³ Language models read and write text in chunks called tokens. In English, a token can be as short as one character or as long as one word.

achieve adequate prompt interpretability by OpenAI models in terms of result format and complete and correct interpretation of all prompt information.

There are various sizes available for pre-trained T5 models [26] based on a Colossal Clean Crawled Corpus (C4), including small (60 million parameters), base (220 million parameters), large (770 million parameters), 3B (3 billion parameters), and 11B (11 billion parameters).

For evaluating the model performance (see section 2.5.1), we used several LLMs published on the HuggingFace hub, such as argilla/notus-7b-v1, epfl-llm/meditron-7b, HuggingFaceH4/zephyr-7b-beta, meta-llama/Llama-2-13b-hf, and togethercomputer/Llama-2-7B-32K-Instruct [38–42].

Table 1 summarizes methods and expected objectives of our study.

2.2 Dataset

The annotated dataset used for all experiments consists of four sources: Bayer Literature Database (PubMed), ADECorpus (PubMed) [28], Social Media Mining for Health Applications (SMM4H, Twitter) [43], and drugs.com[44] and 890 records and is publicly available as supplementary material in [32]. Each record consists of one or several sentences. The dataset was consistently annotated with controlled quality, contains diverse data sources, and the entities are relevant for PV. The dataset is small, diverse, and very clean and allows to train ML models with limited resources. The detailed entity description, data selection, and the annotation process are described in [32].

2.3 Prompting

We performed manual prompt optimization for OpenAI models to obtain the best results based on a few samples of input texts and describe our approach below. Many additional methods exist, and a growing body of literature is discussing additional methods like adding personas or chain-of-thought prompting [45]. We decided to stop spending additional effort in prompt engineering once we reached satisficing results.

We used OpenAI’s prompt engineering guide [46] to achieve better results in terms of adequate prompt interpretability of the text provided. We applied some strategies which are adequate for the given tasks:

- Ask the model to adopt a persona
- Use delimiters to clearly indicate distinct parts of the input
- Instruct the model to answer using a reference text

In contrast to Hu et al. [29], who provided annotation guideline-based prompts and annotated samples via few-shot learning, we intentionally started with zero shot experiments and without any further explanation what is meant by a given entity description (e.g., indication or adverse event) to investigate the native LLM entity extraction capability.

We realized that the task definition in our prompt should be as exact as possible (e.g., three *directly* consecutive hashtags instead of three consecutive hashtags) and preferably with an example (e.g., for product event combination: “Aspirin|Rash”).

Table 1 Overview about the experiments executed and their related objective

Experiment	Objective
GPT 3.5 and GPT 4 prompt engineering	Proper prompt interpretability in terms of result format, completeness, and correct interpretation of all prompt information
GPT reproducibility experiments	Investigating whether GPT provides deterministic output as a prerequisite for computer system validation to demonstrate that a system is suitable for its intended purpose
Open-source LLM selection by executing zero-shot performance and comparison with zero-shot GPT 3.5 and GPT 4	a) Investigation of the performance of 7B and 13B open-source LLMs by using “guided generation” to select the best performing LLM for further performance improvement steps b) Demonstration which entities can be identified by LLMs out-of-the-box c) Investigation which LLM has the best zero-shot performance to be used potentially for generating training data proposals d) Investigation if the expected better performance of GPT 4 versus GPT 3.5 justifies the higher costs and longer execution time per request
Comparison of selected best performing open-source LLM processed (i.e., fine-tuned, few-shot) against zero-shot model, fine-tuned T5, and zero-shot GPT	a) Investigation of potential LLM performance improvement by few-shot learning and QLoRA fine-tuning compared to the zero-shot mode, T5, and GPT b) Proposal for use of LLMs in a regulated environment (as a conclusion of all results)

For all our experiments, the following prompt example was used with the variable part embedded by three hashtags as start and end tag (except for fine-tuned T5 which did not require any prompting). We used all entity labels which are described in [32]:

- **System prompt** (only OpenAI models, system prompt refers to the set of instructions that guide the output): *“I want you to act as an experienced and diligent annotator in a Pharmacovigilance department. Follow the user's instructions carefully. Respond using plain text.”*
- **User prompt** (with an input text example used for all reproducibility experiments, the user prompt is the specific input or question you want the AI to respond to):
- *“Use the following items and identify all items in the text marked by three directly consecutive hashtags (e.g., ###text to be analyzed###) put in front and at the end of the text:*

01. Adverse event
02. Mode of action
03. Administration form / Primary packaging
04. Administration route
05. Comorbidity
06. Dosage
07. Drug / Device
08. Indication
09. Intended effect
10. Medical history / condition
11. Method / Procedure / Administration
12. Outcome
13. Product Dose Combination (PDC)
14. Product Event Combination (PEC)
15. Product Indication Combination (PIC)
16. Product Technical Complaint (PTC)
17. Target parameter
18. Target population

Present the results in the following form:

- *Per item use a new line with the format: 'item: hit(s)'*
- *List multiple hits separated by semi-colon*
- *When you do not find an entry fill the field with 'None'*
- *When you find a product combination (PEC, PIC, PDC) separate Drug / Device from the other Adverse Event, Indication, or Dose term by a pipe character (e.g., 'Aspirin|Rash')*

'Aspirin|Rash')

- *List all items with preceding numbers*
- *Stop generating after the list is complete*

Input text:

“###After TACE for intrahepatic metastasis, localized CCRT (45 Gy over 5 weeks with conventional fractionation and hepatic artery infusional chemotherapy using 5-fluorouracil as a radiosensitizer, administered during the first

and fifth weeks of radiotherapy) was used to treat main HCC with PVT.###”

2.4 Reproducibility Experiments

Reproducibility is of paramount importance for production use of AI models in a regulated environment. This was also acknowledged by the OpenAI developers. In November 2023, two new OpenAI model versions were released addressing this need (gpt-4-1106-preview, gpt-35-turbo-1106). Both models were used for our reproducibility experiments.

In the OpenAI Cookbook [47] the following explanation is provided:

“Reproducibility has always been a big request from user communities when using our APIs ... Developers can now specify seed parameter in the Chat Completion request to receive (mostly) consistent outputs. To help you keep track of these changes, we expose the system_fingerprint field... The system fingerprint is an identifier for the current combination of model weights, infrastructure, and other configuration options used by OpenAI servers to generate the completion.”

The reproducibility of model results (used models: gpt-4-1106-preview, gpt-35-turbo-1106, HuggingFaceH4/zephyr-7b-beta) was evaluated by providing a single record multiple times as the same request ($n = 100$) and analyzing the number of different model responses. The prompting described in section 2.3 was used.

For reproducibility assessment, the number of different responses was counted. In all experiments and all models, the temperature was set to zero and the seed number was omitted or set to 42 to analyze result differences.

2.5 Model Performance

Our motivation for this experiment was to compare the performance of externally hosted OpenAI models versus internally hosted LLMs. In our experiments, we were focusing on partial scores on the positive class (F1 and recall) to evaluate overlapping hits between prediction and truth [32]. We started by running zero-shot performance studies with a range of transformer models to select the best model for further development steps like few-shot prompting or fine-tuning and compared the results of the selected model against two OpenAI models (gpt-4-1106-preview, gpt-35-turbo-1106) as well as a fine-tuned T5.

We assumed that GPT 4 would outperform GPT 3.5, but we also wanted to investigate if the potentially better performance justifies the higher costs and longer execution time per request.

2.5.1 Model Selection

Transformer models were selected, which could be potentially run and fine-tuned with reasonable technical efforts and costs. Therefore, we focused on recent 7B and 13B parameter models which were at the time of selecting LLMs for our study (November / December 2023) ranking high on Hugging Face leader boards [48]: argilla/notus-7b-v1, epfl-llm/meditron-7b, HuggingFaceH4/zephyr-7b-beta, meta-llama/Llama-2-13b-hf, and togethercomputer/Llama-2-7B-32K-Instruct.

Any selection of models will be an extreme reduction compared to the models published on the Hugging Face hub, which hosts, as of June 2024, over 100k models for the Text Generation task. It's common for models to be published in different versions, like meta-llama/Llama-2-13b-hf or meta-llama/Llama-2-13b-chat-hf. When multiple versions of a candidate model were available, we focused on the instruct models in favor of chat models. This was because instruct models tend to perform better in use cases that require adherence to structured output to make use of the generated text. Chat models tend to excuse themselves and are wordy in their outputs, which makes them suitable for use in chat applications but less favorable for data processing tasks.

We compared the model performance against a T5 model fine-tuned on splits of the published dataset [32]. The performance evaluation was executed to identify the candidate model for further processing (few-shot and fine-tuning experiments). We focused a) on the named entities indication, adverse event, drug/device, dose, product event combination, product indication combination, and product dose combination due to business relevance and since the dataset provides the highest number of occurrences for those entities and b) on one data source (ADE Corpus), because the majority of dataset records originated from this source ($n = 500$ of 890 records). All model selection experiments were performed as zero-shot and executed only once ($n = 1$).

2.5.2 Guided Generation

Initial experiments indicated that the prompt for GPT 3.5 and GPT 4 was not generating useful results with the selected 7B or 13B LLMs. The major issue was that open-source models did not adhere to the desired output format and output termination.

Performance of LLMs is highly sensitive to how they are prompted, and a growing body of literature is addressing various techniques to optimize results [45, 46]. One important issue of open-source models is adherence to the expected output format, which is unlikely to be fixed by prompt optimization alone.

One possible option to address the issue with the output format is to use optimized single prompts per entity, but with the disadvantage of additional work and complexity.

We used guided generation (version 0.1.10, 22 December 2023), which offers a guaranteed and stable solution for this issue without the need for additional prompting efforts to compare multiple open-source LLMs with the same user prompt developed for GPT. There are several approaches of controlling the output of LLMs, e.g., [49, 50]. In our paper we evaluated two approaches of guidance-ai for guided generation [50]:

- **Constrained generation to input words:** Enforcing output structure, hard constraints on text generation to words occurring in the input text and in addition, generation termination after newlines. This approach is based on the idea that the output of a NER task can be limited to the input text since generated words which do not occur in the input text are by definition non-valid responses. This constraint works well for NER tasks but breaks down quickly when more free form answers are needed (as required e.g., for relation extraction).
- **Guided generation:** Enforcing output structure and unconstrained generation within free text blocks, with termination of generation after newlines. This approach entirely focuses on adherence to a consistent output structure, which greatly simplifies downstream processing of model inference results and separates the model ability to adhere to a described output format and generating meaningful answers.

Due to the use of product combinations and the use of pipe characters in the expected results, we focused on guided generation for the following analysis. Our experiments with these two approaches showed that imposing known constraints on LLM inference greatly simplifies further usage and processing of LLM results and can bridge the gap between model performance of vastly different parameter sizes.

2.6 T5 and Zephyr Fine-Tuning

T5 was fine-tuned as described in [32] by using the published dataset, executing stratified five-fold cross validation with 80:20 splits. Stratification was performed based on data source and entity type. HuggingFaceH4/zephyr-7b-beta was selected for fine-tuning as we saw the most consistent and promising results from this model. The same dataset as for the T5 models was used for training and testing to create 5-fold cross validation Zephyr models. The quantized low rank adapter (QLoRA) method [51] was used to quantize a pre-trained model to 4-bit and adds a small set of learnable (QLoRA) weights which are

tuned by back propagating gradients through the quantized weights. QLoRA reduces the average GPU memory requirements significantly by using novel high-precision techniques such as 4-bit NormalFloat, double quantization, and paged optimizers without significant performance degradation [51]. We used the X-LLM library for streamlining model training optimization and LLM finetuning [52].

For the fine-tuning of HuggingFaceH4/zephyr-7b-beta, we used an Adam optimizer (paged_adamw_8bit) and set the maximum quantization samples to 1024. The learning rate was set to 0.0002, batch size was set to 1, and the epoch was set to 1. The T5 fine-tuning parameters are described in [32].

2.6.1 Few-Shot Experiments

As a first step towards optimizing model performance, we performed few-shot experiments which were based on the same conditions as for the zero-shot experiments. In total, eight samples (about 1% of dataset) were selected (two samples per data source) which represented per data source the highest number of entities per record. We ensured that “None” was present in our few-shot experiments to teach the models that “None” is a valid option. We executed the experiments with the original HuggingFaceH4/zephyr-7b-beta model and our fine-tuned Zephyr models. For the evaluation we focused on the source ADE Corpus (which contains the most records, $n = 500$).

3 Results

3.1 Prompting

Table 2 shows two examples of GPT 4 model responses representing the two most frequent model results by using system and user prompts introduced in section 2.3 [Input text: “After TACE for intrahepatic metastasis, localized CCRT (45 Gy over 5 weeks with conventional fractionation and hepatic artery infusional chemotherapy using 5-fluorouracil as a radiosensitizer, administered during the first and fifth weeks of radiotherapy) was used to treat main HCC with PVT.”].

Both GPT models created a string as a response. In a post-processing step, we searched for the entity label and used, for comparison of prediction and truth, the information after the colon until the end of line. The T5 procedure is described in [32]. For all other models, we used guided generation and directly created structured data in the same format.

3.2 Reproducibility Experiments

As described in section 2.4, reproducibility experiments were executed to evaluate the number of different responses without a post-processing step on three different models with a temperature of 0.0. The model HuggingFaceH4/zephyr-7b-beta showed the expected behavior, i.e., in case the seed number was omitted we observed different responses ($n = 7$, data not shown), when seed number was set, we observed only one unique response (i.e., $n = 1$, data not shown). Using gpt-35-turbo-1106 and gpt-4-1106-preview we observed while omitting the seed number different results as expected (gpt-35-turbo-1106, $n = 10$, number of different system fingerprints = 1, and gpt-4-1106-preview $n = 33$, number of different system fingerprints = 6, data not shown), but when we set the seed number, GPT 3.5 and GPT 4 did not show a significant reduction in the response variability (gpt-35-turbo-1106, $n = 8$, number of system fingerprints = 1, data not shown, and gpt-4-1106-preview $n = 24$, number of system fingerprints = 4). In Figure 1, the GPT 4 distribution of different responses and the number of system fingerprints per response is shown. About 1/3 of all system responses ($n = 33$ of 100) contain the same contents but distributed over all four available system fingerprints (see Fig. 1, left blue bar and left red bar). In Fig. 2 the total number and the unique number of responses is displayed. Nearly half of all responses (blue bar, $n = 48$ of 100) are related to one system fingerprint and represent about 44 % of all unique responses (red bar, $n = 20$ of 45). Figures 1 and 2 demonstrate that different fingerprints can generate the same result, and one fingerprint can generate different responses. It is obvious that the established OpenAI concept of system fingerprint and seed number provides only insufficient capabilities to achieve consistent reproducibility. As a consequence of our results, we will not invest in further performance improvements of GPT 3.5 and 4 since in their current state they are not fit for application in a regulated environment. We do not claim this to be a discovery and are not speculating about potential explanations for this behavior.

We do not see any inherent reason why these models could not be technically made reproducible with technical or legal efforts. Especially since this issue does not occur when running models locally and controlling for reproducibility, as our application of open models show in other parts of this paper.

3.3 Measurement of Model Performance

3.3.1 Zeroshot Experiments

Figures 3 and 4 show the partial F1 and partial recall of selected 7B and 13B LLMs for seven entities for all models under consideration and by using the same prompt.

Table 2 Two GPT 4 examples are shown to demonstrate the responses selected from the highest and second highest numbers of total response shown in Fig 1. The differences between both results are shown in bold faces

	Response 1	Response 2	Truth
1	01. Adverse event: none	01. Adverse event: none	01. Adverse event: none
2	02. Mode of action: none	02. Mode of action: none	02. Mode of action: radiosensitizer
3	03. Administration form/Primary packaging: none	03. Administration form/primary packaging: none	03. Administration form/primary packaging: None
4	04. Administration route: hepatic artery infusional chemotherapy	04. Administration route: hepatic artery infusional chemotherapy	04. Administration route: hepatic artery infusional chemotherapy
5	05. Comorbidity: intrahepatic metastasis	05. Comorbidity: intrahepatic metastasis	05. Comorbidity: intrahepatic metastasis, PVT
6	06. Dosage: 45 Gy over 5 weeks; during the first and fifth weeks of radiotherapy	06. Dosage: 45 Gy over 5 weeks; during the first and fifth weeks of radiotherapy	06. Dosage: 45 Gy over 5 weeks; during the first and fifth weeks of radiotherapy
7	07. Drug/device: 5-fluorouracil	07. Drug/device: 5-fluorouracil	07. Drug/device: 5-fluorouracil
8	08. Indication: main HCC with PVT	08. Indication: main HCC with PVT	08. Indication: intrahepatic metastasis; main HCC with PVT
9	09. Intended effect: none	09. Intended effect: none	09. Intended effect: chemotherapy, treat
10	10. Medical history/condition: none	10. Medical history/condition: none	10. Medical history/condition: none
11	11. Method/procedure/administration: TACE; localized CCRT; conventional fractionation	11. Method/procedure/administration: TACE; CCRT; hepatic artery infusional chemotherapy	11. Method/procedure/administration: TACE; localized CCRT; conventional fractionation; hepatic artery infusional chemotherapy; radiotherapy
12	12. Outcome: none	12. Outcome: None	12. Outcome: none
13	13. Product dose combination (PDC): 5-fluorouracil 45 Gy over 5 weeks	13. Product dose combination (PDC): 5-fluorouracil 45 Gy over 5 weeks	13. Product dose combination (PDC): 5-fluorouracil 45 Gy over 5 weeks
14	14. Product event combination (PEC): none	14. Product event combination (PEC): None	14. Product event combination (PEC): none
15	15. Product indication combination (PIC): 5-fluorouracil main HCC with PVT	15. Product indication combination (PIC): 5-fluorouracil main HCC with PVT	15. Product indication combination (PIC): 5-fluorouracil intrahepatic metastasis; 5-fluorouracil main HCC with PVT
16	16. Product technical complaint (PTC): none	16. Product technical complaint (PTC): none	16. Product technical complaint (PTC): none
17	17. Target parameter: none	17. Target parameter: none	17. Target parameter: none
18	18. Target population: none	18. Target population: none	18. Target population: none

Gy, Gray (SI unit); HCC, hepatocellular carcinoma; PVT, portal vein thrombosis; TACE, transarterial chemoembolization; CCRT, concurrent chemoradiotherapy; PDC, product dose combination; PEC, product event combination; PIC, product indication combination; PTC, product technical complaint

Fig. 1 Distribution of gpt-4-1106-preview responses (blue) and of system fingerprints (red) observed by providing 100 times the same request. The bars are sorted by unique total response count in descending order

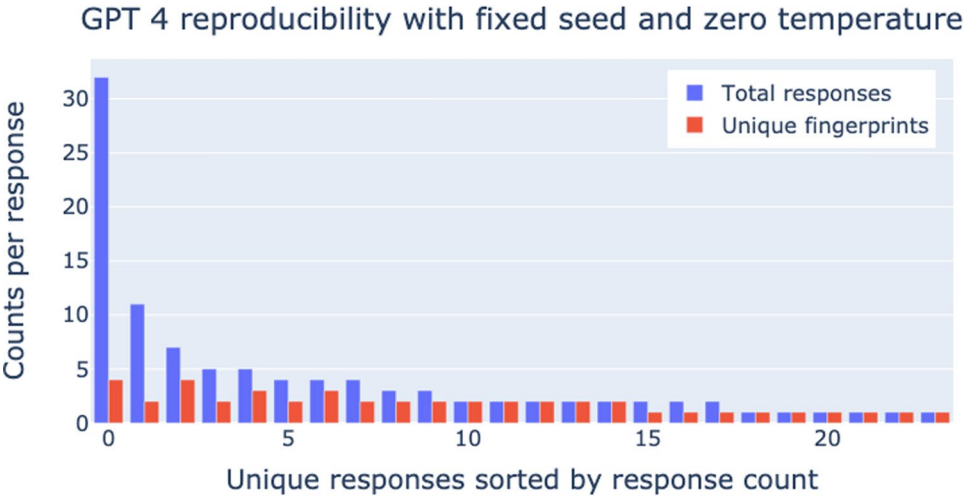


Fig. 2 Distribution of gpt-4-1106-preview total (blue) and unique responses (red) observed per system fingerprint. The bars are sorted by the total response count per fingerprint in descending order

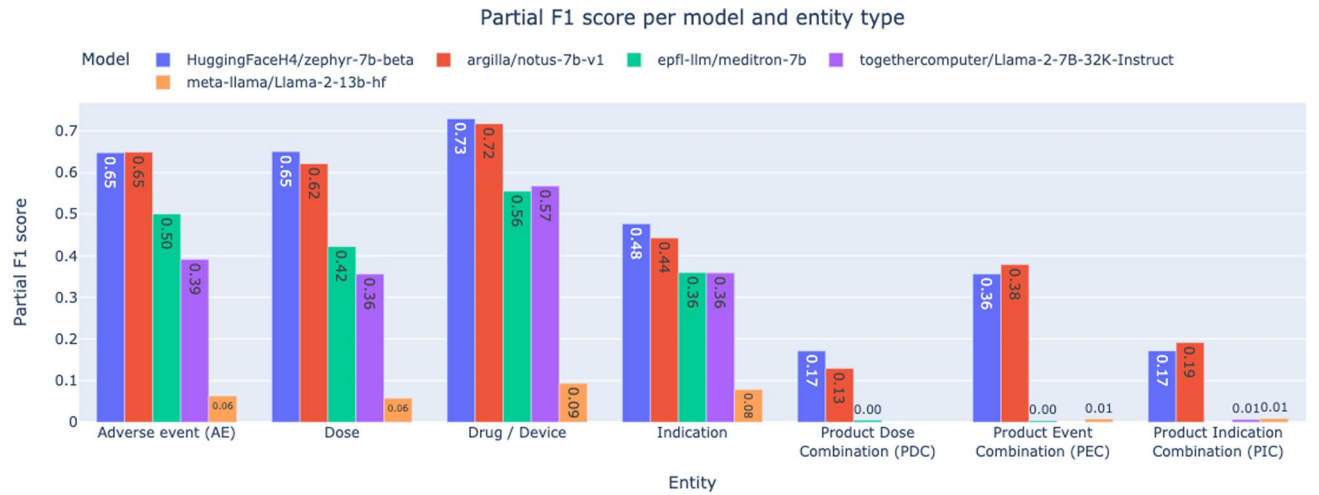
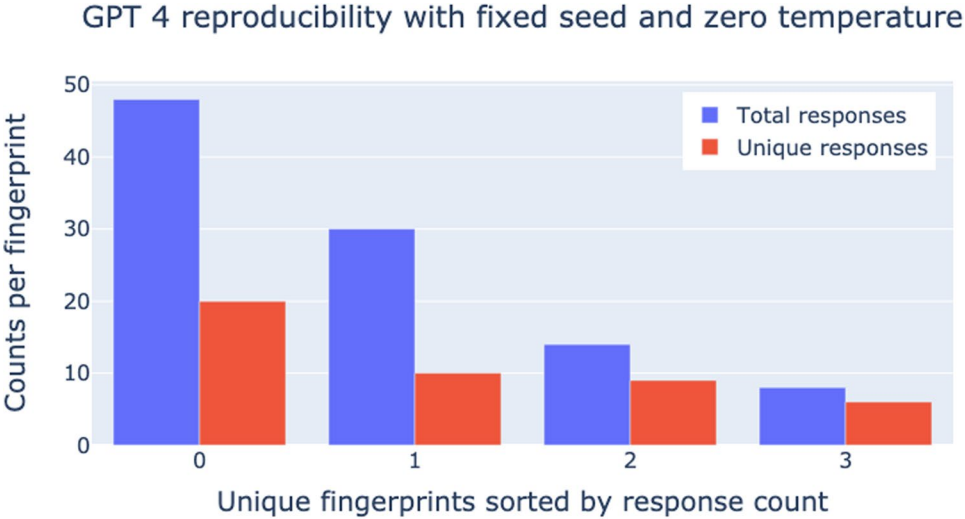


Fig. 3 Partial F1 entity performance of open-source LLMs. *AE*, adverse event; *PDC*, product dose combination; *PEC*, product event combination; *PIC*, product indication combination

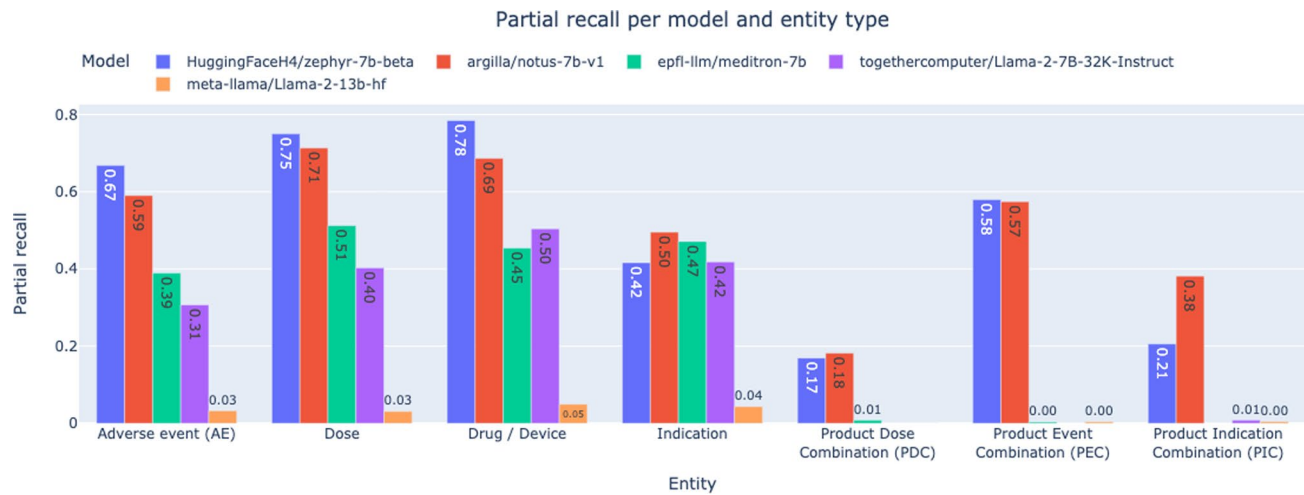


Fig. 4 Partial recall entity performance of open-source LLMs. *AE*, adverse event; *PDC*, product dose combination; *PEC*, product event combination; *PIC*, product indication combination

HuggingFaceH4/zephyr-7b-beta and argilla/notus-7b-v1 performed best compared to the other three open-source LLMs. The meta-llama/Llama-2-13b-hf performed significantly worse than the other LLM models. As shown in Fig. 4, the Zephyr model in our experiment performs better regarding partial recall than the Notus model, and therefore, was selected for further investigation (i.e., fine-tuning and few-shot inference).

3.3.2 Comparison Between Fine-Tuned Zephyr and Fine-Tuned T5 Models

Figure 5 shows the partial F1 and partial recall results of the fine-tuned T5 and fine-tuned Zephyr models. The partial F1 score of a fine-tuned T5 is higher compared with Zephyr for all selected entities. Especially for product combinations, T5 performs much better than Zephyr with the given prompt.

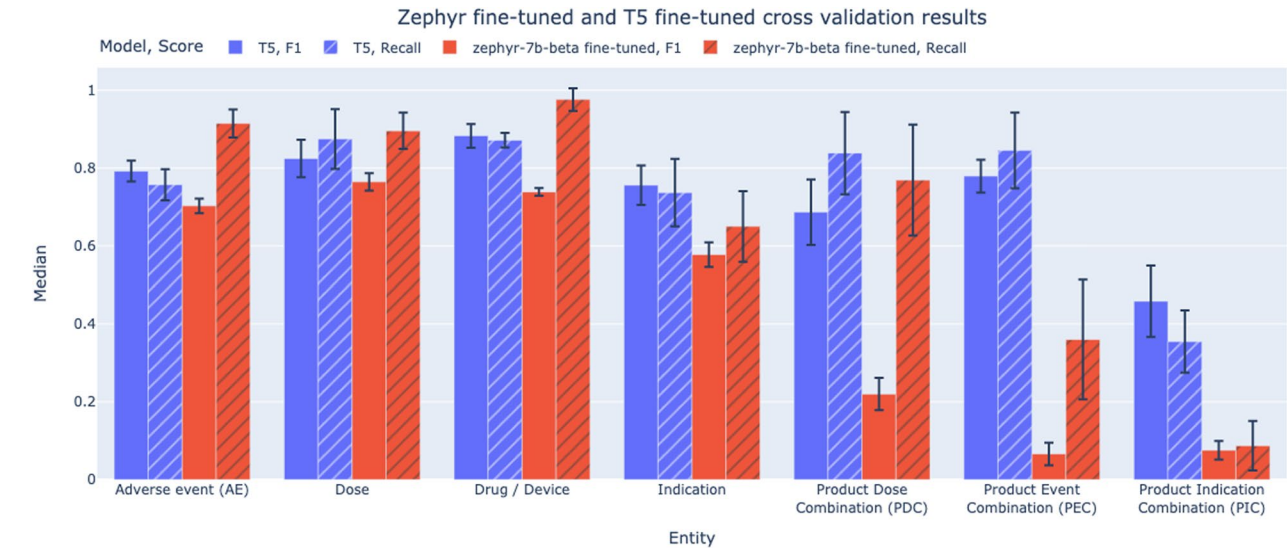


Fig. 5 Partial F1 score and recall results (median and standard deviation) of entities from stratified cross-validation with fine-tuned T5 and fine-tuned Zephyr models. *AE*, adverse event; *PDC*, product dose

combination; *PEC*, product event combination; *PIC*, product indication combination

3.3.3 Comparison of Fine-Tuned T5 with Few-Shot Pre-trained and Fine-Tuned Zephyr

Figure 6 shows the partial F1 score of few-shot experiments for seven selected entities by comparing the zero-shot Zephyr model with fine-tuned Zephyr and fine-tuned T5 models. Please note that zero-shot experiments were executed once ($n = 1$), while each model of the cross-validated fine-tuned model types was tested ($n = 5$). The fine-tuned Zephyr model shows slightly higher F1 scores compared to the zero-shot Zephyr model and when referring to the non-overlapping standard deviations, the performance of T5 may be considered as significantly higher than with fine-tuned

Zephyr. Independent of any possible prompt optimization, we assume the reason for the better performance observed may originate from the fact that in contrast to fine-tuned Zephyr all T5 weights were adjusted by fine-tuning. Please note that we did not execute a complete adjustment of all Zephyr weights since the efforts and computation costs would be significantly increased due to the vast differences in model parameters (T5 Base: 220 million parameters versus 7 billion parameters).

In Figs. 7 and 8 the median partial F1 scores and partial recall of all zero- and few-shot experiments are displayed. Figures 7 and 8 demonstrate that the scores for non-product combination entities of GPT 4 and T5 are similar, while the

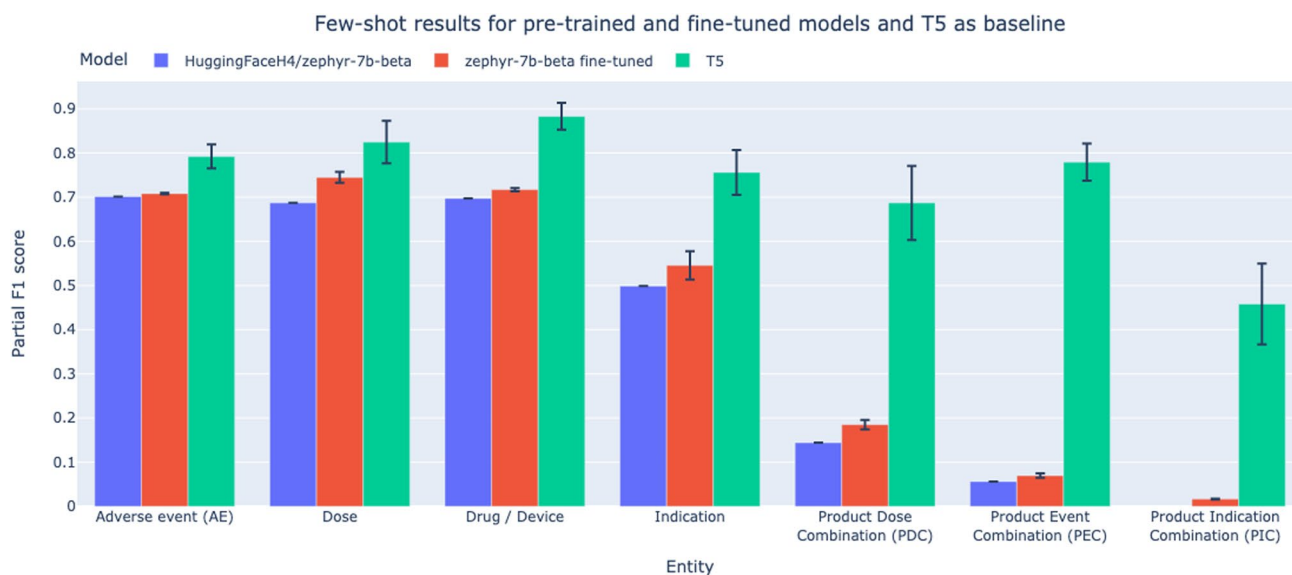


Fig. 6 Few-shot median partial F1 score of fine-tuned T5 and Zephyr compared with zero-shot Zephyr. Standard deviation is shown for all fine-tuned models. *AE*, adverse event; *PDC*, product dose combination; *PEC*, product event combination; *PIC*, product indication combination



Fig. 7 Comparison of median partial F1 scores of fine-tuned T5 and Zephyr versus zero-shot Zephyr, GPT 3.5, and GPT 4. *AE*, adverse event; *PDC*, product dose combination; *PEC*, product event combination; *PIC*, product indication combination; *zephyr-FT*, fine-tuned Zephyr-7b-beta

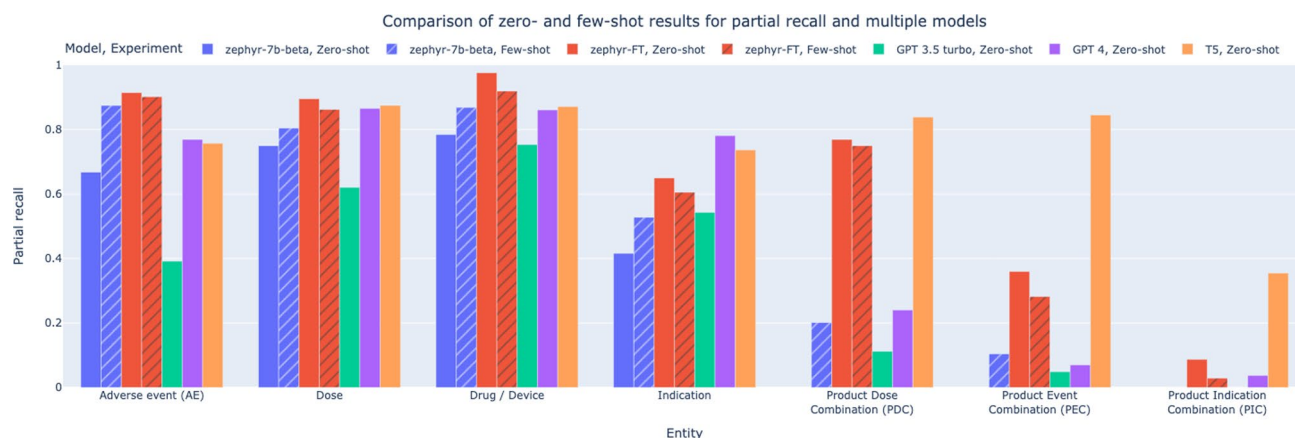


Fig. 8 Comparison of median partial recall of fine-tuned T5 and Zephyr versus zero-shot Zephyr, GPT 3.5, and GPT 4. *AE*, adverse event; *PDC*, product dose combination; *PEC*, product event combination; *PIC*, product indication combination; *zephyr-FT*, fine-tuned Zephyr-7b-beta

scores for product combination entities of T5 are significantly higher compared to all other models. The F1 score of AE detection with GPT 4 is significantly higher than with GPT 3.5. The recall of the fine-tuned Zephyr model with regard to AE, drug/device, and dose seems to perform slightly better than the scores of other models. Comparing zero-shot Zephyr with GPT 3.5, the Zephyr AE F1 and recall performance is better. Figure 8 demonstrates that the recall performance can be increased even by a small number of samples ($n = 8$) in few-shot learning.

4 Discussion

4.1 Selection of the Best Performing Open-Source LLM

In section 3.3.1 we investigated the performance of selected LLM models [53–55] in zero-shot experiments. We selected models with seven and thirteen billion parameters which allows to run and fine-tune those models with reasonable technical efforts and costs. We could demonstrate that a higher parameter count (number of model parameters) does not necessarily imply a better performance for a given NER task.

Parameter count is an obvious but alluring proxy to predict model performance on unseen tasks, which can be summarized as the “bigger is better” heuristic [56]. This heuristic holds badly if applied too broadly but can be a good guiding principle when applied in a narrower sense where one focuses on models that are based on the same general architecture, similar training data, similar training tasks, as well as objective functions.

We demonstrated that model parameter count is not a suitable proxy metric for downstream task performance.

As shown in this paper, by considering our study framework as prompts and text generation techniques, the LLMs investigated were not able to outperform a relatively small model by today's standards, such as T5 for the task discussed here. This only considers task performance; once one factors in that smaller models greatly simplify the complexity and effort to develop and maintain the overall system, the benefit of small local models starts to become more and more significant in many industry-relevant settings.

One example of this observation is that a popular open model like LLAMA-2 13B can be outperformed by smaller models based on similar architecture when the training data is chosen carefully. Especially instruction tuning, such as for Zephyr-7b-beta, can efficiently bridge the gap in model size.

Figure 3 shows two models with a similar architecture, such as Llama-2-7B-32K-Instruct and Llama-2-13b-hf, but different pre-training data, where the smaller model outperforms the bigger one, but the bigger one was not trained with instruction tuning using reinforcement learning with human feedback (RLHF). This highlights that various methods of instruction tuning can increase model performance on unseen tasks since they shape the trained model in a way that focuses less on next token or masked token predictions, but more on fulfilling a task that is given to the model in the form of text input. Kirstain et al. demonstrated that in open question answering tasks, enlarging the training set does not improve performance. In contrast, classification, extractive question answering, and multiple choice tasks benefit so much from additional examples that collecting a few hundred examples is often “worth” billions of parameters [57].

As shown in Figs. 3 and 4, the Zephyr model performs better than all other open-source models, and therefore, was selected for further investigation (i.e., fine-tuning and few-shot inference)

Zephyr-7b-beta was derived from a strong baseline LLM Mistral-7B [58] and was further fine-tuned with the aim of fulfilling tasks using direct preference optimization (DPO) as a novel and promising fine-tuning approach [59]. We are using in our study a small, but well curated dataset [32]. We fine-tuned a small T5 model with this dataset, which shows in our study an excellent performance for designated NER tasks (see Figs. 6, 7, 8). This is in line with recent publications which demonstrates that remarkably strong performance can be achieved from only a handful of examples in the training data [32, 60].

4.2 Comparison of Zephyr, GPT 3.5, and GPT 4 Zero-Shot Performance

In comparing the OpenAI models, GPT 4 shows a better NER performance in general in Fig. 7, but only the GPT 4 AE F1 score is significantly higher than the GPT 3.5 score. As shown in Fig. 8, the better recall performance of GPT 4 is even more pronounced. Both GPT and Zephyr models demonstrate good NER zero-shot performance, while Zephyr shows a very good recall performance. Zero-shot GPT performance of named entity linking with product combination entities could be improved by few-shot learning as shown in relation extraction experiments [27]. AE detection is considered as a more difficult task for LLMs compared with other entities used in this study (as drug, dose, or indication) [32]. For complex NER tasks, the significant performance increase would likely justify the higher GPT 4 costs and longer execution time per request compared with GPT 3.5. For AE detection, zero-shot Zephyr demonstrates better performance than GPT 3.5 and could alternatively be used.

4.3 Comparison of Best Performing Open-Source LLM (Zero-Shot, Few-Shot, Fine-Tuned) with Fine-Tuned T5, and Zero-Shot GPT

With the given prompt, we observed similar F1 performances of zero-shot GPT 4 and fine-tuned T5 for non-product combination entities, while the product combination scores of T5 are significantly higher compared with all other models. The results imply that compared with a small T5-Base with complete weight fine-tuning performance, QLoRA fine-tuned models could not achieve comparable performance especially with regard to production combination entities. To improve non-T5 models, further efforts in prompt engineering are possible (e.g., few-shot techniques or prompt optimization). All quantitative results regarding model performance are lower bounds and must be interpreted as such. They do not inform about a potential performance ceiling that could be achieved with more efforts on prompting or fine-tuning. Zero-shot experiments

with minimal prompting efforts are an adequate measure to predict how well a model will work with minimal effort.

In Figs. 7 and 8 we found that the F1 performances of zero-shot, few-shot, and fine-tuned Zephyr model show similar results, but the recall performance of fine-tuned Zephyr improved significantly. We demonstrated that a zero-shot Zephyr model performance could be increased by fine-tuning and few-shot learning, even by using a very small sample ($n = 8$).

The same data were used for fine-tuning T5 and fine-tuning Zephyr. When comparing fine-tuned Zephyr with T5, T5 had a better F1 performance, especially with product combination entities.

All approaches used in this paper offer potential for additional gains in task performance, and a balance needs to be found between time spent on specific approaches. From the experiments we performed, we expect that few-shot approaches offer the potential for additional performance gains with relatively little additional investment, which could be addressed in future research.

The technical content of this study is concerned with named entity recognition and linking (i.e., product combination entities), but the conclusions drawn here may hold in principle for many more use cases in the text modality, which is due to the new paradigm opened by LLMs. Traditionally, different use cases (e.g., sentiment classification, question answering) were approached using different methods, but LLMs allow casting any use case that operates in a modality covered by a model as a text generation task based on a prompt and input data.

4.4 Proposal for Use of LLMs in a Regulated Environment

Translating such results into qualified applications running as regulated systems will prove challenging, and several hurdles are needed to be overcome. One such hurdle is the reproducibility of model inference, which is discussed in this paper. Reproducibility is a cornerstone of a computer validation process [21–24], as it allows to independently verify and validate the documented findings of previous validation runs. In our study we investigated the overall performance and reproducibility capability of the responses produced by LLMs, regarding NER conducted upon PV texts. We demonstrated in section 3.2 that in contrast to a fine-tuned Zephyr, recent OpenAI model implementations failed in showing reproducible results, which is a prerequisite for computer system validation to demonstrate that a system is suitable for its intended purpose.

Our results clearly show that this point needs special attention, which might include and is not limited to legal efforts and technical considerations. Given the time at which the experiments were done and presented here, it is evident

that the current level of publicly available GPT 4 and 3.5 models do not provide the level of reproducibility needed in these contexts.

Depending on the provider of an external LLM, one might also need to consider the additional need to keep a validated model in a controlled state and available over longer time periods. Additional documentation of the training data and inference logic might also be useful for developing sufficient levels of trust in the model's performance on new and unseen data.

When models are running internally, one can control hardware as well as inference code and can technically ensure reproducibility. Open models [61] can additionally offer better understanding in terms of training data and procedures used for model training and offer paths towards the explainability of specific model inferences.

Running inference locally on LLMs is demanding in terms of hardware requirements such as CPU inference speed or available GPU memory. Today's data centers and cloud providers offer the capability to run LLMs from a few billion to much larger number of parameters, but the complexity and costs increase steeply with the number of model parameters. Currently, models in the range of 7B to 13B are in a reasonable spot of potentially sufficient capabilities and simplicity of operation, system development, and experimentation.

Depending on the task and available experience, even in smaller organizations, one can be confident that open models may be the most reasonable choice for many tasks. This is especially true when dedicated domain-relevant prior knowledge is deployed within the overall system of model inference and additional components.

Out-of-the-box and without changing prompts, smaller open models show a big capability gap compared with state-of-the-art hosted models such as GPT 4 and GPT 3.5. Our analysis clearly showed that for this task, the gap can be closed using guided generation and fine-tuning with modest amounts of training data. Creating this training data are costly and need to be factored in when developing AI systems. However, using hosted models does not mean one can safely develop a system without the generation of training data, since it is needed to evaluate and monitor model performance. Data-efficient training regimes emphasize that the combination of open and smaller models with dedicated task-specific data can reach and outperform closed models' performance.

Generating training data requires domain expertise, adherence to well-described labeling protocols, and measures for quality control like inter-annotator agreement. Time and cost requirements of generating high-quality data can potentially be reduced by using the best available hosted model to generate data candidates, which are mostly quality

controlled by domain experts. Our findings and those of other publications clearly demonstrate that LLMs possess the capability to accomplish this task [27, 29, 30]. This approach shows that initially opposing ideas of open versus closed and internal versus external models are not that relevant for many practical applications.

5 Conclusion

LLM-based AI systems have great innovative potential for applications in the pharmaceutical industry. Our case study shows that, notwithstanding the current rate of progress, general rules of system implementation in a controlled environment have not lost their importance. At the center of developing a pharmacovigilance system, efficacy (i.e., system performance metrics), reproducibility of results, and operability (i.e., to remain a validated system in a controlled status) should be the focus. For system deployment a robust technical and legal guardrail framework needs to ensure that models remain in a controlled status and provide availability over time, as well as reproducible model inference, as minimum criteria for them to be deployed operationally. These issues aside, they could be used to greatly reduce the effort to generate proposals for fine-tuning data or automated system evaluation and monitoring. This is clear from the fact that GPT 3.5 and 4 showed promising results with limited prompting efforts in zero-shot settings. Hence, their results could replace annotating efforts to generate training data needed to train bespoke models like T5.

Here we took the traditional route of using data labeled by domain experts to ensure high data quality, as it is most suitable for a regulated environment. We were unable to reach sufficient levels of reproducible model inference with GPT 3.5 and GPT 4 models in contrast to locally maintained open-source models, which implies that for the current GPT versions an independent verification in computer system validation of previous validation runs is hardly to achieve.

Smaller model sizes offer great advantages in effort and complexity when developing and maintaining an LLM-based AI system. It seems plausible that model size and the size of fine-tuning data could be traded off to reach the required levels of system performance. However, we have not shown this explicitly here.

Based on this case study, we recommend using the model which performs best for NER and relation extraction (e.g., GPT), irrespective of reproducibility to generate proposals for data annotation or to generate synthetic data for fine-tuning. Then, one should fine-tune, evaluate, and deploy the most operationally feasible LLM that reaches the required level of performance on the task to find the optimal balance

of complexity, stability, and quality for the overall system in a regulated environment.

The key contributions of our paper to the research community are as follows:

- GPT 3.5 and GPT 4, as of today, should not be used as a part of a GxP validated system due to missing guarantees of reproducibility.
- Due to promising results of Zephyr or GPT in terms of NER tasks tested in our study it seems in many relevant cases to be sufficient to use only zero-shot experiments to generate training data for further manual review.
- This training data could be used to train a smaller, e.g., T5 Base system. This would be a straightforward process to train a high-quality and resource-efficient model, which could be used as a part of a GxP system.

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Declarations

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Conflict of interest Jürgen Dietrich holds shares in Bayer AG. Jürgen Dietrich is a full-time employee of Bayer AG. André Hollstein holds shares in Bayer AG. André Hollstein is a full-time employee of Bayer AG. Jürgen Dietrich and André Hollstein have no conflict of interest that are directly relevant to the content of this experiment. The views expressed in this paper are those of the authors and do not necessarily reflect the official policy or position of Bayer AG.

Ethics approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

Availability of data and material The dataset used is publicly available in publication [32] under Supplementary file 1, https://static-content.springer.com/esm/art%3A10.1007%2Fs40264-023-01322-3/MediaObjects/40264_2023_1322_MOESM1_ESM.xlsx

Code availability The code used for this experiment is not provided, as a patent is pending.

Authors' contributions Jürgen Dietrich and André Hollstein were involved in the conception and design of the experiments and result interpretation. André Hollstein was responsible for code generation of guidance AI implementation and the execution of few-shot experiments, Jürgen Dietrich was responsible for the rest of the code generation tasks and the execution of cross-validation and LLM finetuning experiments. All authors contributed to the interpretation of the data analysis results and assisted with the concept and draft revisions of the manuscript. All authors reviewed and approved the final manuscript and accept full responsibility for its overall content.

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